Individual Task 1: Part 1

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This report examines a Data Scientist role at the Australian Council for Educational Research (ACER) and investigates two education datasets using machine learning. Support Vector Machine (SVM) and Random Forest models are used to identify patterns in student outcomes and compare their predictive performance.

Overleaf project URL: https://www.overleaf.com/project/6a7c7a2c4ad5779049ceef04

1 Data Science Role

1.1 Job Description and Role Identification

The Australian Council for Educational Research (ACER) is looking for a Data Scientist to work in its Data Science and Automation Program [1]. The role involves using data science and machine learning to build solutions for assessment, reporting and research. The job also includes working with different types of data, testing and monitoring models, and considering privacy, fairness and explainability.

Job listing URL: https://au.seek.com/job/93511797

1.2 Industry Context

ACER works in education research and assessment. Data science can help analyse student performance, improve assessments and reporting, find patterns in education data, and support better decisions. These methods can help education organisations understand student outcomes and develop practical ways to improve assessment and learning.
[Word count: 44]

1.3 Role Focus

The role involves both finding useful information in data and making predictions. The job mentions machine learning, NLP, analytical problems, AI-enabled products, and testing and monitoring models. This shows that the role is not only about analysing past data but also about using data to support future decisions.
[Word count: 48]

1.4 Value Creation

The role creates value by turning education data into useful tools and solutions. These can help ACER with assessment, reporting and research. Using responsible AI is also important because the solutions need to be reliable, explainable and careful with people's data. This can make the results more useful to ACER and its stakeholders.
[Word count: 49]

1.5 Values and Culture Alignment

I value learning new things, solving practical problems and working with others, which matches the type of environment described by ACER. I also like using technology to solve real problems. ACER's work in education and its focus on responsible AI match my interest in using data science in a useful and responsible way.
[Word count: 48]

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1.6 Diversity and Inclusion

People from different backgrounds can bring different ideas and experiences to a data science project. This is important in education because students have different backgrounds and circumstances. A diverse team can help identify bias, question assumptions and make the data and models more representative and fair.
[Word count: 45]

1.7 Cover Letter

The tailored cover letter is provided as a stand-alone letter in the Appendix.

2 Data Set

The first dataset is the UCI Student Performance dataset, containing 649 students and 30 features covering demographic, social, school and academic information [4, 5]. The second is the UCI Predict Students' Dropout and Academic Success dataset, containing 4,424 students and 36 features covering demographic, academic and socioeconomic information [6, 7]. Both datasets are available from the UCI Machine Learning Repository.

Dataset 1: https://archive.ics.uci.edu/dataset/320/student+performance

Dataset 2: https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success

I selected these datasets because they relate closely to the ACER role. They allow me to examine student academic performance and longer-term student outcomes using machine learning.
[Word count: 91]

3 Data Analysis

3.1 Approach

Two machine learning models, Support Vector Machine (SVM) and Random Forest, were used to analyse the two education datasets [2, 3]. The data was split into 80% training and 20% testing data. SVM features were standardised, while categorical variables in the Student Performance dataset were converted into numerical variables.

For the Student Performance dataset, students with a final grade (G3) of 10 or above were treated as Pass and students below 10 were treated as Fail. This assumption allowed the dataset to be analysed as a binary classification problem.

3.2 Results and Insights

For the Student Dropout and Academic Success dataset, Random Forest performed slightly better than SVM. Random Forest achieved 76.84% accuracy and a 75.42% F1-score, compared with 75.82% accuracy and a 74.54% F1-score for SVM.

The classification results also showed differences between student outcome groups. Both models identified Graduate students well, with 93% recall. However, both models struggled with the Enrolled group, where recall was only 35% for SVM and 36% for Random Forest. This suggests that overall accuracy does not show how well the models perform for every student group.

The Random Forest model relied most on features related to academic progress. The most important features included approved curricular units and grades from the first and second semesters. These features were useful for prediction, but they should not be interpreted as direct causes of student outcomes.

3.3 Model Performance

Table 1 summarises the performance of the two models across both datasets.

Table 1. Model performance

Dataset	Model	Accuracy	Precision	F1-score
Dropout	SVM	75.82%	74.65%	74.54%
Dropout	Random Forest	76.84%	75.48%	75.42%
Performance	SVM	77.69%	90.10%	86.26%
Performance	Random Forest	80.00%	85.00%	88.70%

For the Student Performance dataset, Random Forest again performed better overall, reaching 80.00% accuracy and an 88.70% F1-score. However, its confusion matrix showed an important limitation: it correctly identified only 2 of the 20 students who failed. SVM identified 10 of the 20 failing students. This shows that a higher overall accuracy does not necessarily mean that a model performs well for every group.

For this dataset, the Random Forest model gave the greatest importance to previous failures, followed by school type and intention to continue to higher education. Absences and parental education were also among the more important features. These features helped the model make predictions, but they should not be treated as causes of student outcomes.

3.4 Evaluation and Implications

Accuracy was used to compare overall prediction performance, while precision and recall provided more detail about the types of predictions made. F1-score was also used because it balances precision and recall. These measures are standard classification evaluation measures available in scikit-learn [8, 9]. Confusion matrices were also examined to identify errors affecting particular student groups.

Overall, Random Forest was slightly stronger across both datasets, but neither model performed equally well for every outcome. The two datasets provided complementary insights: the first focused on academic performance, while the second focused on student progression and academic outcomes.

The results suggest that machine learning could support educational decision-making, but predictions should be treated as supporting information rather than final decisions. Further validation, fairness checks and regular monitoring would be needed before using such models in a real educational setting.

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Generative AI Attribution

This work was primarily human-created. AI was used to make stylistic edits, such as changes to structure, wording, and clarity. AI was also used for content-related support, including explaining concepts, troubleshooting code, and helping organise ideas. AI was prompted for its contributions, and AI-generated content was reviewed and approved by the author. The following model/application was used: ChatGPT.

I certify that the submission of this assignment is incomplete without the submission of the completed **Condition 3 Bounded Process AI Declaration Form**.

Acknowledgments

I acknowledge the Traditional Custodians of the lands on which RMIT University operates and pay my respects to their Elders past and present.

References

[1] Australian Council for Educational Research. 2026. Data Scientist. SEEK job advertisement. https://au.seek.com/job/93511797 Accessed: 14 August 2026.

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[5] Paulo Cortez and Alice Maria Goncalves Silva. 2008. Using Data Mining to Predict Secondary School Student Performance. In *Proceedings of the 5th Annual Future Business Technology Conference*. EUROSIS, Porto, Portugal, 5–12. https://archive.ics.uci.edu/dataset/320/student+performance

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[7] Valentim Realinho, Monica Vieira Martins, Jorge Machado, and Luis Baptista. 2021. Predict Students' Dropout and Academic Success. doi:10.24432/C5MC89 Accessed: 14 August 2026.

[8] Scikit-learn developers. 2026. classification_report. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.classification_report.html Accessed: 14 August 2026.

[9] Scikit-learn developers. 2026. Metrics and Scoring: Quantifying the Quality of Predictions. https://scikit-learn.org/stable/modules/model_evaluation.html Accessed: 14 August 2026.

A Cover Letter

Kathan Shah
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13 August 2026

To the
Hiring Manager
Australian Council for Educational Research (ACER)
Melbourne, VIC

Re: Data Scientist Position

Dear Hiring Manager,

I am writing to apply for the Data Scientist position at the Australian Council for Educational Research (ACER). I am interested in this role because it combines data science, machine learning and real-world education problems. I would value the opportunity to use my technical skills in an organisation where data can contribute to better assessment, research and education outcomes.

I completed a Bachelor of Engineering in Computer Engineering and have developed practical experience in Python, machine learning and data science through academic projects and internships. During my internships, I worked on machine learning and Python-based projects, which gave me experience with data preparation, model development and solving practical problems using data. These experiences have also helped me become comfortable learning new tools and working through technical problems independently.

The responsibilities of this position particularly interest me because they involve machine learning, analytical problems, data products and responsible use of AI. I also value the role's focus on privacy, fairness, explainability and collaboration. I believe these areas are important when working with educational data because decisions based on data can affect students and other members of the education community.

I would bring a willingness to learn, a practical approach to problem-solving and an interest in applying technology to meaningful problems. I am particularly attracted to ACER's research-driven environment and its focus on using data and technology to improve education.

Thank you for considering my application. I would welcome the opportunity to discuss how my background and interest in data science could contribute to ACER.

Yours sincerely,

Kathan Shah

B GitHub Repository

GitHub repository: https://github.com/kathanaus/individual-task-1-data-science/tree/main